

Lecture Notes: CNN-1

Understanding padding, stride, and output feature-map dimensions

Symbols and Definitions

For a convolutional layer, we use the following notation.

Symbol	Meaning
N	Input size in one dimension, for example length in 1D.
H_{in}	Input image height.
W_{in}	Input image width.
K	Kernel or filter size in one dimension.
K_h, K_w	Filter height and filter width.
P	Padding size added to both sides in one dimension.
P_h, P_w	Padding added along height and width.
S	Stride, meaning how many cells the filter moves each step.
S_h, S_w	Stride along height and width.
C_{out}	Number of output filters, also equal to the number of output channels.

Why Is the Padded Size $N + 2P$?

Consider a 1D input of length N :

$$[x_1, x_2, x_3, \dots, x_N]$$

If the padding is P , then we add P cells on the left and P cells on the right. Therefore, the total number of extra cells is:

$$P + P = 2P$$

So the padded input size becomes:

$$N_{\text{padded}} = N + 2P$$

Small 1D Example

Suppose:

$$N = 5, \quad P = 1$$

Original input:

$$[x_1, x_2, x_3, x_4, x_5]$$

After adding one padding cell on the left and one on the right:

$$[0, x_1, x_2, x_3, x_4, x_5, 0]$$

The new size is:

$$1 + 5 + 1 = 7$$

which matches:

$$N_{\text{padded}} = 5 + 2(1) = 7$$

What About Corners in 2D Padding?

For a 2D image, padding is added around the full border. If the original image has size:

$$H \times W$$

and padding P is applied equally on all sides, then:

- P rows are added at the top;
- P rows are added at the bottom;
- P columns are added at the left;
- P columns are added at the right.

Therefore, the padded height is:

$$H_{\text{padded}} = H + 2P$$

and the padded width is:

$$W_{\text{padded}} = W + 2P$$

So the padded image size is:

$$(H + 2P) \times (W + 2P)$$

Corner Example

Suppose the original image is 3×3 :

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

With padding $P = 1$, we add zeros around the full border:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & a & b & c & 0 \\ 0 & d & e & f & 0 \\ 0 & g & h & i & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The output padded image is 5×5 because:

$$H_{\text{padded}} = 3 + 2(1) = 5$$

$$W_{\text{padded}} = 3 + 2(1) = 5$$

The corner cells are included automatically in the padded rows and padded columns. They are not an extra separate term. This is why we do not write something like $H + 2P + 4$ or $W + 2P + 4$.

Proof of the 1D Convolution Output Size

Now we prove the feature size formula from the possible filter positions.

Assume a 1D input with original length N . After padding P on both sides, the length becomes:

$$N_{\text{padded}} = N + 2P$$

Let the filter size be K . The filter must fit fully inside the padded input.

First Valid Filter Position

The first valid starting index is:

$$0$$

Last Valid Filter Position

The filter has size K . If it starts at position j , then it covers:

$$j, j+1, j+2, \dots, j+K-1$$

The last covered index must be at most the final index of the padded input:

$$N + 2P - 1$$

Therefore:

$$j + K - 1 \leq N + 2P - 1$$

Subtracting $K - 1$ from both sides gives:

$$j \leq N + 2P - K$$

So the last valid starting position is:

$$N + 2P - K$$

Stride and Number of Positions

If the stride is S , then the possible starting positions are:

$$0, S, 2S, 3S, \dots, mS$$

The last valid starting position must satisfy:

$$mS \leq N + 2P - K$$

Dividing by S :

$$m \leq \frac{N + 2P - K}{S}$$

The largest integer value of m is:

$$m = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor$$

Since the counting starts at 0, the number of positions is:

$$m + 1$$

Therefore, the 1D convolution output size is:

$$O = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

This O is the number of places where the filter can be applied. Each valid placement produces one output value.

2D Convolution Output Size

A 2D image has height and width. We apply the same 1D logic separately to the height direction and to the width direction.

For height:

$$H_{\text{out}} = \left\lfloor \frac{H_{\text{in}} + 2P_h - K_h}{S_h} \right\rfloor + 1$$

For width:

$$W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} + 2P_w - K_w}{S_w} \right\rfloor + 1$$

Therefore, the spatial output feature-map size is:

$$H_{\text{out}} \times W_{\text{out}}$$

If the convolutional layer has C_{out} filters, then the full output tensor size is:

$$H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}}$$

Worked Example

Suppose the input image size is:

$$32 \times 32$$

The convolution layer uses:

$$K_h = K_w = 3, \quad P_h = P_w = 1, \quad S_h = S_w = 1$$

For height:

$$H_{\text{out}} = \left\lfloor \frac{32 + 2(1) - 3}{1} \right\rfloor + 1$$

$$H_{\text{out}} = \lfloor 31 \rfloor + 1 = 32$$

For width:

$$W_{\text{out}} = \left\lfloor \frac{32 + 2(1) - 3}{1} \right\rfloor + 1$$

$$W_{\text{out}} = 32$$

So the spatial feature-map size is:

$$32 \times 32$$

If there are 64 filters, then the full output tensor is:

$$32 \times 32 \times 64$$

Why Padding Can Preserve Spatial Size

In the previous example, the input was 32×32 and the output spatial size was also 32×32 . This happened because a 3×3 filter with padding $P = 1$ and stride $S = 1$ preserves height and width.

For one dimension:

$$O = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

Using $K = 3$, $P = 1$, and $S = 1$:

$$O = \lfloor N + 2 - 3 \rfloor + 1$$

$$O = N - 1 + 1 = N$$

So the output size equals the input size.

Common Mistakes

1. **Forgetting the +1.** The filter positions start at index 0, so the number of positions is $m + 1$, not just m .
2. **Adding corners separately.** Corners are already included when the padded height and padded width are computed.
3. **Using the number of filters as spatial size.** The number of filters controls output channels, not height or width.
4. **Ignoring stride.** Stride controls how many positions are skipped between filter placements.

Summary

Main formula for 1D convolution:

$$O = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

Main formula for 2D convolution:

$$H_{\text{out}} = \left\lfloor \frac{H_{\text{in}} + 2P_h - K_h}{S_h} \right\rfloor + 1$$

$$W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} + 2P_w - K_w}{S_w} \right\rfloor + 1$$

Full output tensor size:

$$H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}}$$

Practice Questions

1. Input size: 28×28 , kernel size: 3×3 , padding: 0, stride: 1, filters: 16. Find the output tensor size.
2. Input size: 64×64 , kernel size: 5×5 , padding: 2, stride: 1, filters: 32. Find the output tensor size.
3. Input size: 32×32 , kernel size: 3×3 , padding: 1, stride: 2, filters: 64. Find the output tensor size.
4. Explain in one sentence why the corner padding cells are not counted as an extra term beyond $H + 2P$ and $W + 2P$.

Answers to Practice Questions

1. $26 \times 26 \times 16$.
2. $64 \times 64 \times 32$.
3. $16 \times 16 \times 64$.
4. Because the corners are already part of the added padded rows and padded columns.